phase2 system design

1. Objective of Phase 2

Upgrade system from:

“all users equal (or simple reliability)”

to:

user-aware credibility using behavior + history

2. What Phase 2 Adds

Core Additions

Time-decayed reliability Experience score (Exp_i) Basic anomaly detection (multi-signal, but lightweight) Improved user weight w_i

Still NOT included

Graph trust (T_i) ML models Location system Memory similarity

3. Updated User Model

3.1 Time-Decayed Reliability

Formula

R_i(t) =

With Decay

alpha_i ← alpha_i * exp(-lambda_r * delta_t) beta_i ← beta_i * exp(-lambda_r * delta_t)

Then update:

if vote correct: alpha_i += 1 else: beta_i += 1

3.2 Experience Score

Formula

Exp_i(t) =

Simplified Computation

E_i += 1 per interaction Exp_i = log(1 + E_i) / log(1 + E_max)

3.3 Basic Anomaly (Lightweight)

Use only 2 signals initially:

1. Rate Deviation

rate = interactions_last_10min baseline = avg interactions per hour D_rate = rate / baseline

2. Entropy (Simplified)

p_up = (#upvotes / total) p_down = (#downvotes / total) H = - (p_up log p_up + p_down log p_down) D_entropy = 1 - (H / log(2))

3.4 Final Anomaly

Anom_i = 1 - (- (*1 D*{rate} + *2 D*{entropy}))

4. Updated User Weight

Phase 2 Weight

w_i = R_i(t) Exp_i(t) (1 - Anom_i)

Interpretation

Reliable × Experienced × Non-suspicious

5. Updated Post Computation

Same as MVP but using new w_i:

Interaction Mass

N_j(t) = w_i e^{-(t - t_k)}

Credibility

C_j(t) =

Variance

Use same simplified version.

6. Updated Alert Condition

Phase 2 Alert

Alert(j) = (C_j > 0.75 ;; N_j > N_{min} ;; Var_j < ^2)

7. Backend Changes Required

User Table (Updated)

users (user_id TEXT PRIMARY KEY, alpha FLOAT, beta FLOAT, experience FLOAT, anomaly FLOAT, last_active TIMESTAMP)

New Computation Step

After each vote:

update_user(i) update_post(j)

8. Pipeline (Phase 2)

Vote received ↓ Update user: reliability experience anomaly ↓ Compute user weight w_i ↓ Update post: N_j, S_j+, S_j- ↓ Compute: C_j, Var_j ↓ Check alert

9. What Phase 2 Achieves

Compared to MVP:

Feature MVP Phase 2 user equality yes no spam resistance low medium new user influence high controlled anomaly detection none basic

10. What You Should Test

Critical Experiments

1. Spam Burst

Many fake users vote quickly

Expect:

Anomaly ↑ → weight ↓ → credibility stable

2. New User Attack

new accounts voting

Expect:

low Exp_i → low influence

3. Honest Users

consistent correct votes

Expect:

R_i ↑ → credibility improves

11. Hyperparameters (Phase 2 Minimal Set)

lambda_r → reliability decay lambda → interaction decay alpha1 → anomaly weight (rate) alpha2 → anomaly weight (entropy) E_max → experience normalization theta → credibility threshold

12. What NOT to Overcomplicate

Avoid:

too many anomaly signals complex ML perfect tuning

13. Phase 2 Summary

MVP: “what users say” Phase 2: “who is saying it matters”

Next Phase (Phase 3 Preview)

You will add:

Graph trust propagation (T_i) Coordination detection Advanced anomaly signals

This is where system becomes attack-resistant at scale.

Final Insight

You now transition from:

basic system → adaptive system